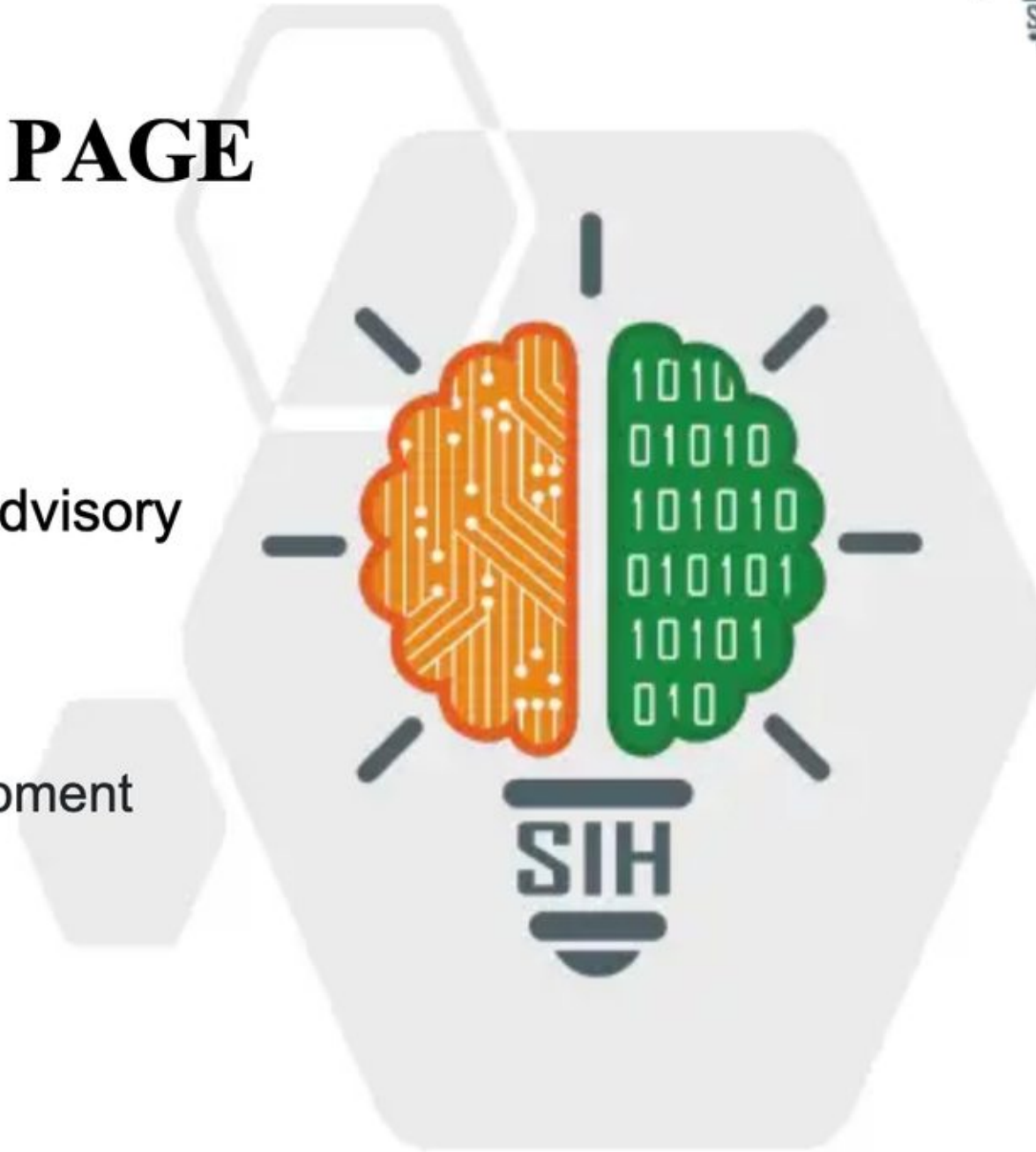


TITLE PAGE

- **Problem Statement ID** – SIH25010
- **Problem Statement Title** - Smart Crop Advisory System for Small and Marginal Farmers.
- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category-** Software
- **Team ID-**
- **Team Name** - Krishi Mitra

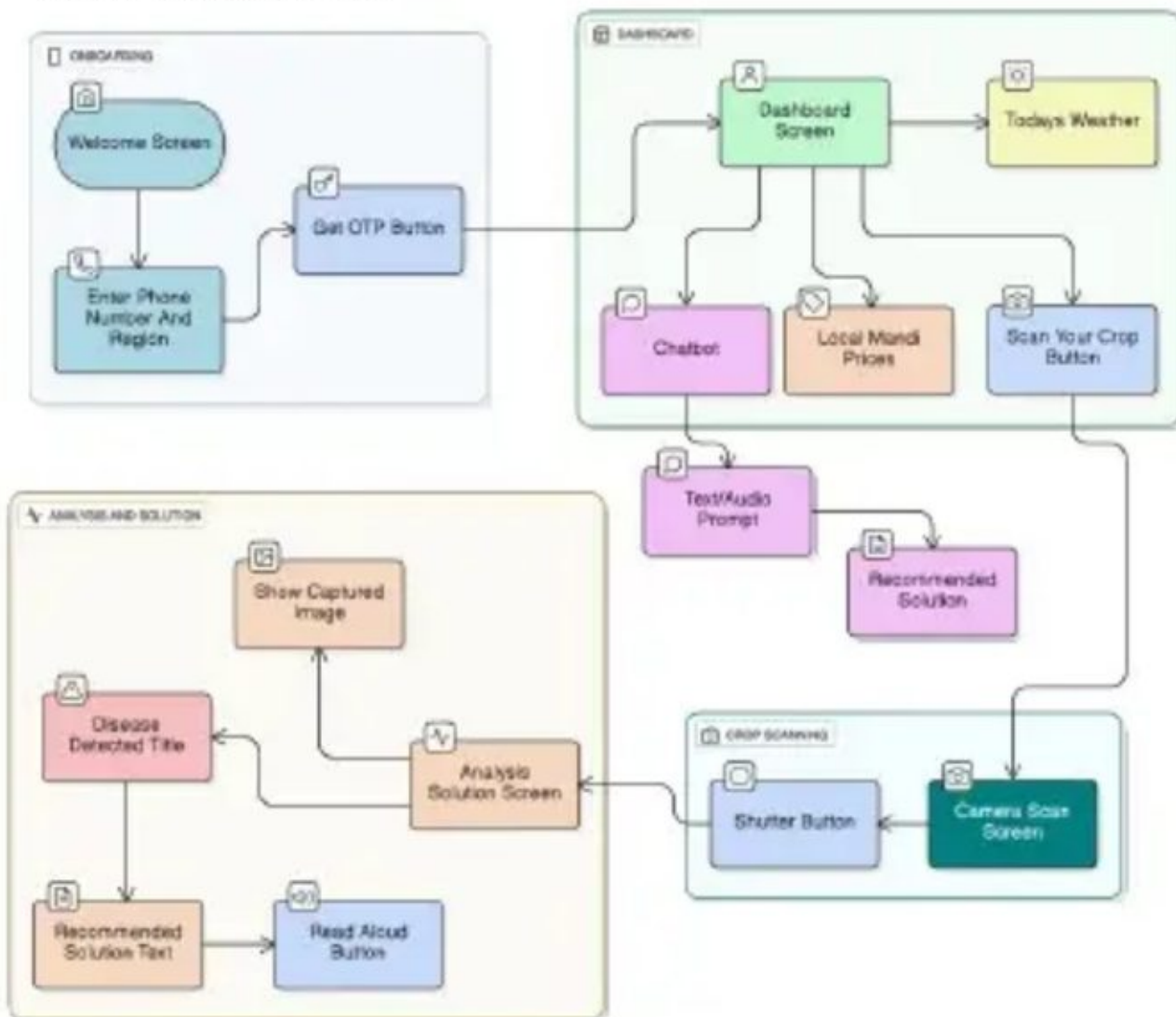




Krishi Mitra: The Farmer's AI Companion



Krishi Mitra User Flow



Proposed Solution (What is it?)

- An AI-powered, multilingual mobile app designed as a 360° digital assistant for Indian farmers.
- Features a simple, **voice-assisted interface** to overcome literacy and technology barriers.
- **Core Modules:**
 - **Snap & Solve:** Image recognition for instant pest and **disease detection**.
 - **Smart Advisory:** **Personalized crop**, fertilizer, and soil health recommendations.
 - **Real-time Alerts:** Location-based weather warnings and daily task lists.
 - **Market Connect:** **Live Mandi prices**, **input cost tracking**, and a platform to connect with buyers.

How it Addresses the Problem

- **Replaces Guesswork:** Provides scientific, data-driven advice instead of unreliable sources.
- **Breaks Barriers:** Offers support in local languages and through **voice commands**.
- **Reduces Costs:** Optimizes the use of fertilizers and pesticides, **preventing waste**.
- **Boosts Income:** Improves yield and provides **market price transparency** to ensure fair sales.

Innovation and Uniqueness

- **Holistic Ecosystem:** Integrates everything from soil preparation to sales in one app.
- **Hyper-Local Focus:** Tailors all advice (weather, crops, market prices, govt. schemes) to the farmer's specific location.
- **Inclusive Design:** Built for low connectivity with **offline features** and a voice-first interface.



Framework:

Flutter (preferred for speed + cross-platform Android/iOS) or React Native.

UI/UX Layer:

- >Material Design / Cupertino Widgets for intuitive navigation.
- >Multilingual support with **i18n libraries**.
- >Voice-first design → large buttons, minimal text.

Offline-first Capability:

- >Local SQLite/NoSQL cache.
- >Service Workers & background sync for syncing once internet is available.

Voice Interaction:

- >**Speech-to-Text** (local + Google Speech API).
- >**Text-to-Speech** for advisories in local languages.



Core Backend:

Python(Flask/FastAPI/Django REST Framework).

API Gateway:

REST/GraphQL endpoints for secure farmer-app ↔ server communication.

Database Layer:

- >**Relational**: PostgreSQL/MySQL for structured farmer profiles, crop data, transactions.
- >**NoSQL**: Firebase/Firestore for real-time alerts, chat logs, notifications.

Caching Layer:

Redis for speeding up repeat queries like weather or mandi prices.

Authentication:

Firebase Auth/OTP-based login for low-literacy users.

Scalability:

Containerized microservices with Docker + Kubernetes (GKE/EKS).



Image Recognition:

Offline → TensorFlow Lite (MobileNetV2, EfficientNet) **trained** on PlantVillage & custom data.

Online → Cloud Vision APIs (Google/AWS) for complex cases.

Natural Language Processing (NLP):

- >Regional language understanding → HuggingFace/IndicNLP.
- >Offline voice queries → Vosk/Coqui STT.

Recommendation Engine:

- >Rule-based + ML hybrid.
- >**Personalized crop & soil advisory using Decision Trees + historical farmer data.**

Predictive Analytics:

- >Time-series models (Prophet, ARIMA) for yield & price forecasting.
- >Weather-driven risk models for irrigation/fertilizer decisions.



Weather:

OpenWeatherMap API, IMD (Indian Meteorological Department) data sources.

Market Prices:

Agmarknet.gov.in API for mandi prices; fallback to crowdsourced farmer data.

Government Schemes: DigiLocker + API sets for subsidy/benefit verification.

Payment Integration (future roadmap):

UPI/Paytm for direct buyer-seller transactions.

GIS & Maps:

Google Maps API / Mapbox for location tagging of farms.

Notifications:

Firebase Cloud Messaging + SMS fallback (for non-smartphone farmers).



Cloud Platform:

Google Cloud / AWS / Azure for scalability, AI APIs, and serverless functions.

CI/CD:

GitHub Actions or Jenkins for automated testing & deployment.

Monitoring & Logging:

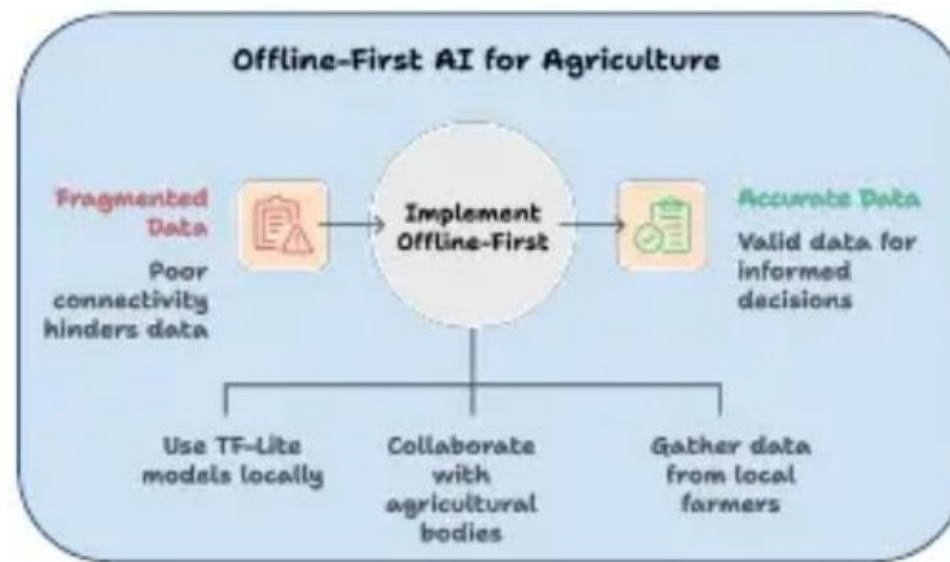
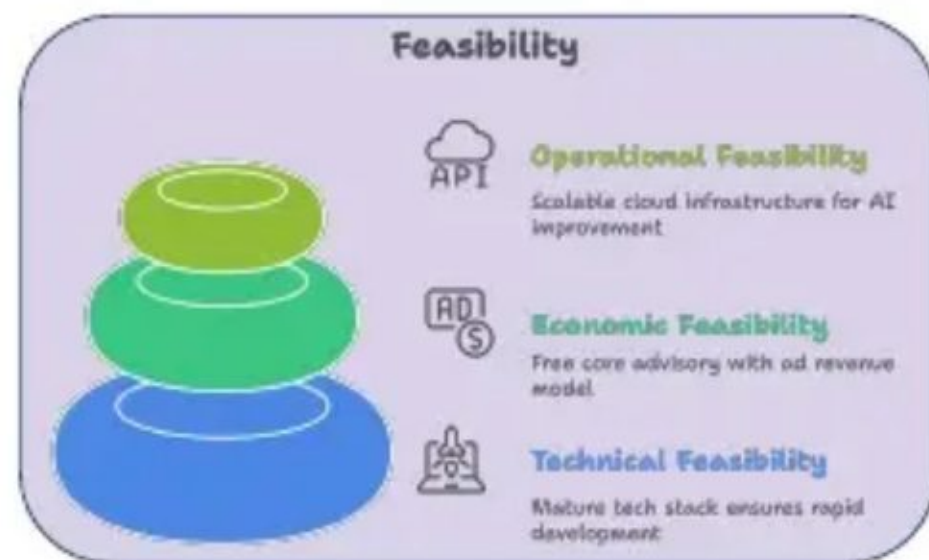
Prometheus + Grafana for health; ELK Stack for data insights.

Data Security:

AES-256 encryption; Blockchain ledger (future) for digital passports.

Scalability:

Kubernetes pods for horizontal scaling; load balancers for high traffic.



Impacts:



Benefit by the idea:

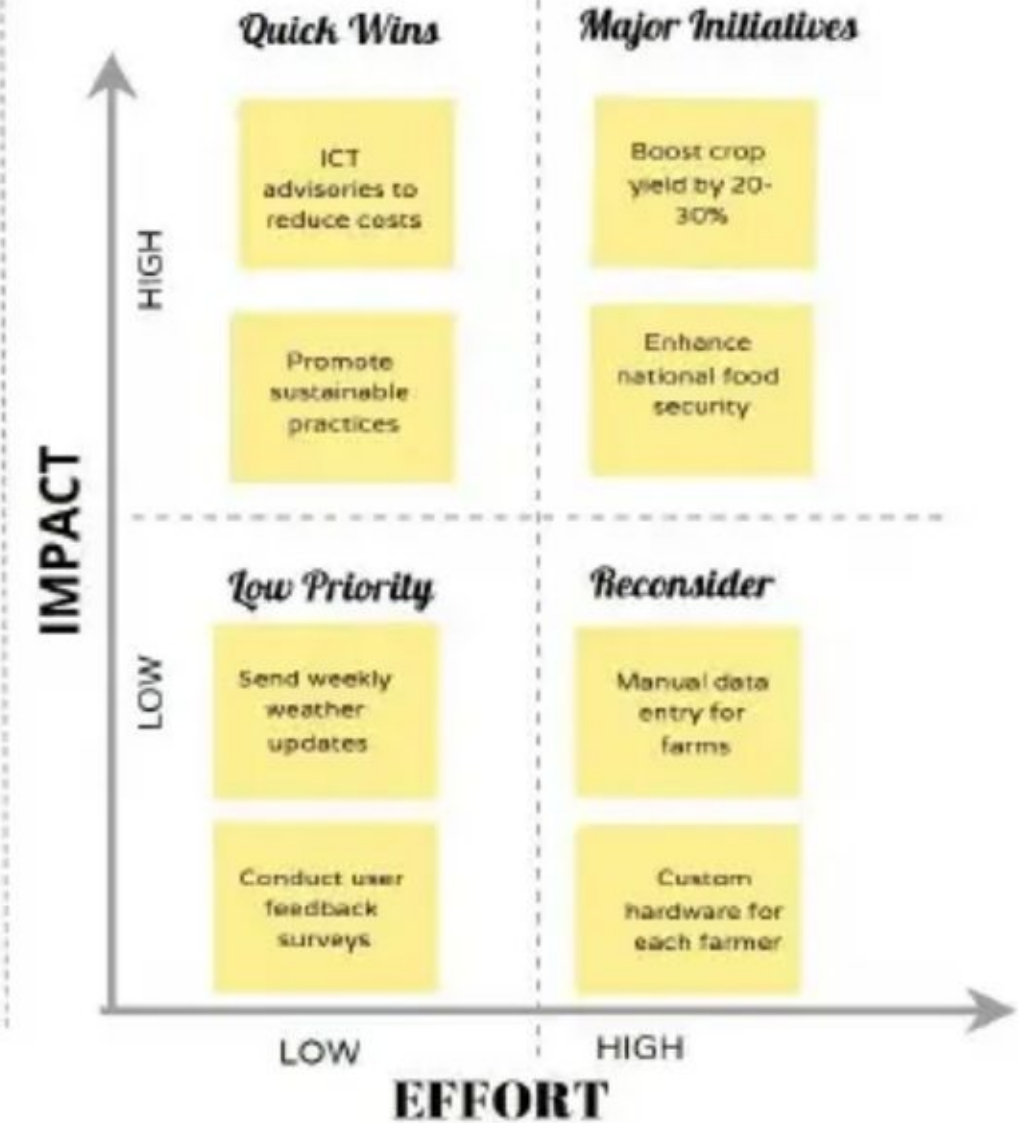
Pest and Disease Management: By using image recognition, the app can identify pests and diseases early on, allowing for targeted treatments that minimize crop damage and reduce the need for widespread pesticide use.

Increased Crop Yield and Quality: AI can analyze data to provide recommendations for optimal planting, irrigation, and harvesting times, leading to healthier and more abundant crops.

Soil and Water Management: AI-powered sensors can monitor soil moisture and nutrient levels, enabling the app to recommend precise irrigation and fertilization strategies. This conserves water and prevents the overuse of fertilizers.

Potential Initiatives

- Increased Productivity
- Reduced Costs
- Higher Income & Empowerment
- Social Benefits
- Environmental Benefits





RESEARCH AND REFERENCES



1. Foundational Data & Reports

- a. [NABARD All India Rural Financial Inclusion Survey \(NAFIS\) Report, 2022](#): To establish the problem's scale and target audience.
- b. [ICAR & KVK Publications](#): For sourcing scientifically validated agricultural practices (e.g., fertilizer dosage, pest control methods).
- c. [FAO \(Food and Agriculture Organization\) Reports](#): For global and national data on food security and sustainable agriculture.
- d. [Using Deep Learning for Image-Based Plant Disease Detection](#)
- e. [A survey on deep learning techniques for plant disease Detection](#)
- f. [A Dataset for Visual Plant Disease Detection and Recognition](#)

2. Technical Resources & Datasets

- a. **Machine Learning Models:**
 - i. **TensorFlow Hub**: For accessing pre-trained models like MobileNetV2 for image classification.
 - ii. **GitHub Repositories**: Search for open-source crop disease detection projects.
- b. **Datasets for AI Training:**
 - i. **PlantVillage Dataset**: A large, public dataset of 54,000+ images of healthy and diseased crop leaves. This is a key resource for training your model.
 - ii. [data.gov.in](#): Search for specific datasets like "District-wise Soil Fertility Maps" or "Crop Production Statistics".
- c. **API Documentation:**
 - i. [agmarknet.gov.in](#): Official source for Mandi (market) price data APIs.
 - ii. **OpenWeatherMap API**: Documentation for accessing real-time and forecasted weather data.

3. Design & Implementation Guides

- a. **Material Design Guidelines**: For building an intuitive and accessible user interface, which is critical for your target audience.
- b. **TensorFlow Lite Implementation Guides**: Official documentation on how to prepare and deploy .tflite models in a Flutter application.